

CS57300
PURDUE UNIVERSITY

DATA MINING

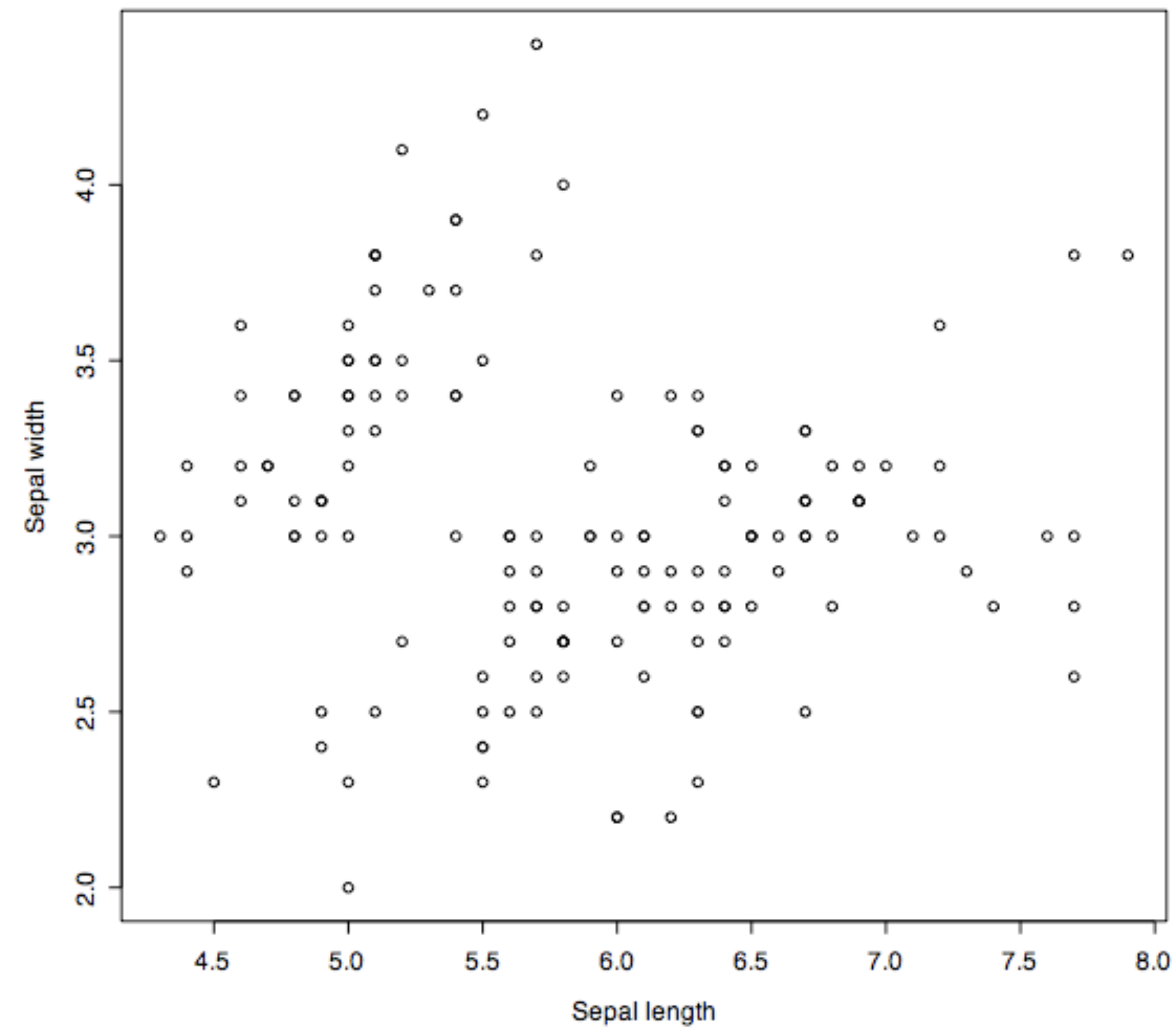
DATA VISUALIZATION

- ▶ Serve for different purposes
 - ▶ Composition: e.g., see for a discrete dimension x_i , the fraction of each values
 - ▶ Distribution: e.g., see the distribution of a continuous dimension of data x_i
 - ▶ Comparison:
 - ▶ Compare values of two continuous dimensions of the data, x_i and x_j
 - ▶ Given discrete x_i , compare the values of x_j when x_i takes different values.
 - ▶ Relationship: e.g., examine the relationship between x_i and x_j

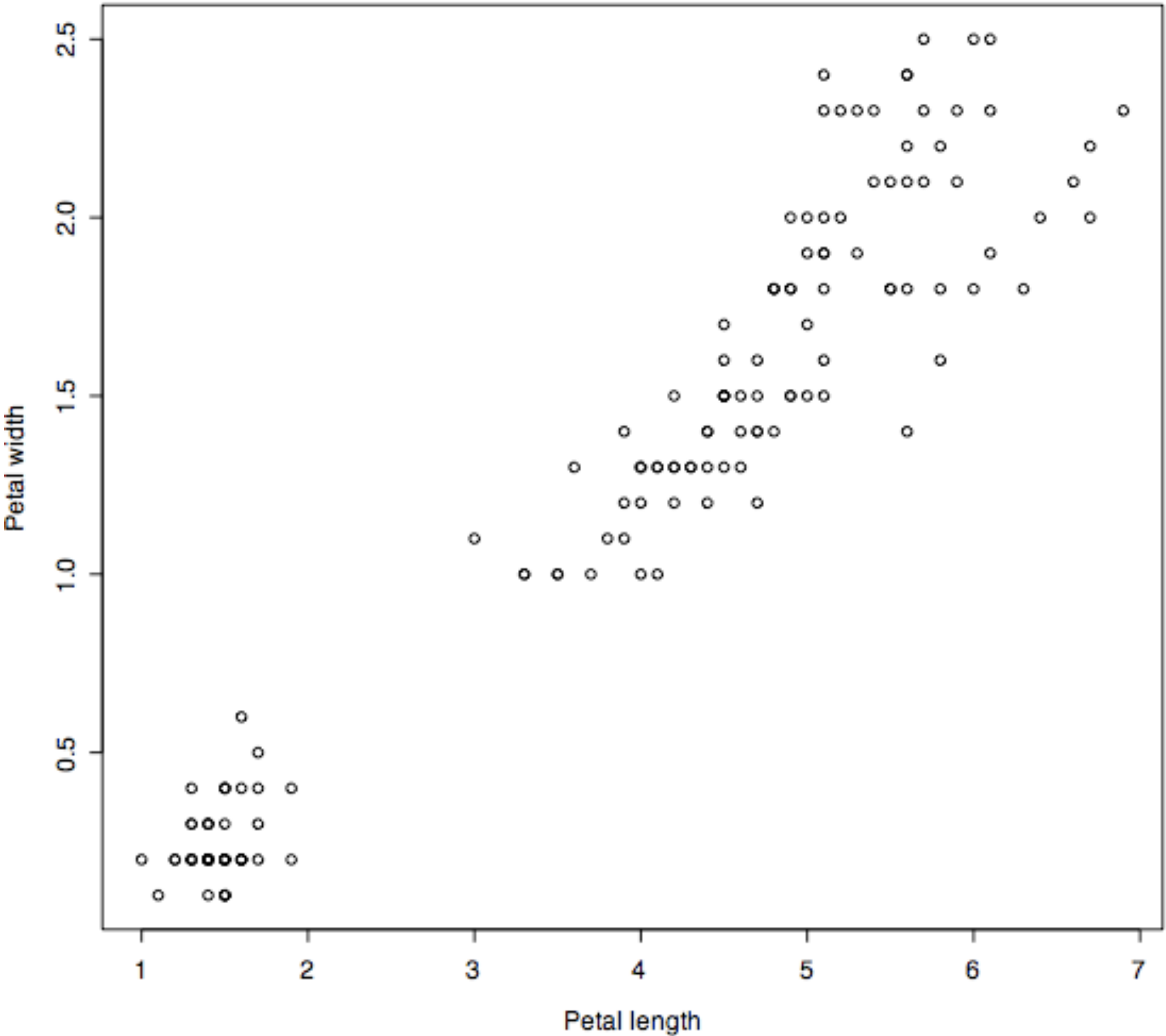
RELATIONSHIP: SCATTER PLOT (2D)

- ▶ Most common plot for bivariate data
 - ▶ Horizontal X axis: the suspected **independent** variable
 - ▶ Vertical Y axis: the suspected **dependent** variable
- ▶ Graphically shows:
 - ▶ If X and Y are related
 - ▶ Linear or non-linear relationship
 - ▶ If the variation in Y depends on X
 - ▶ Outliers

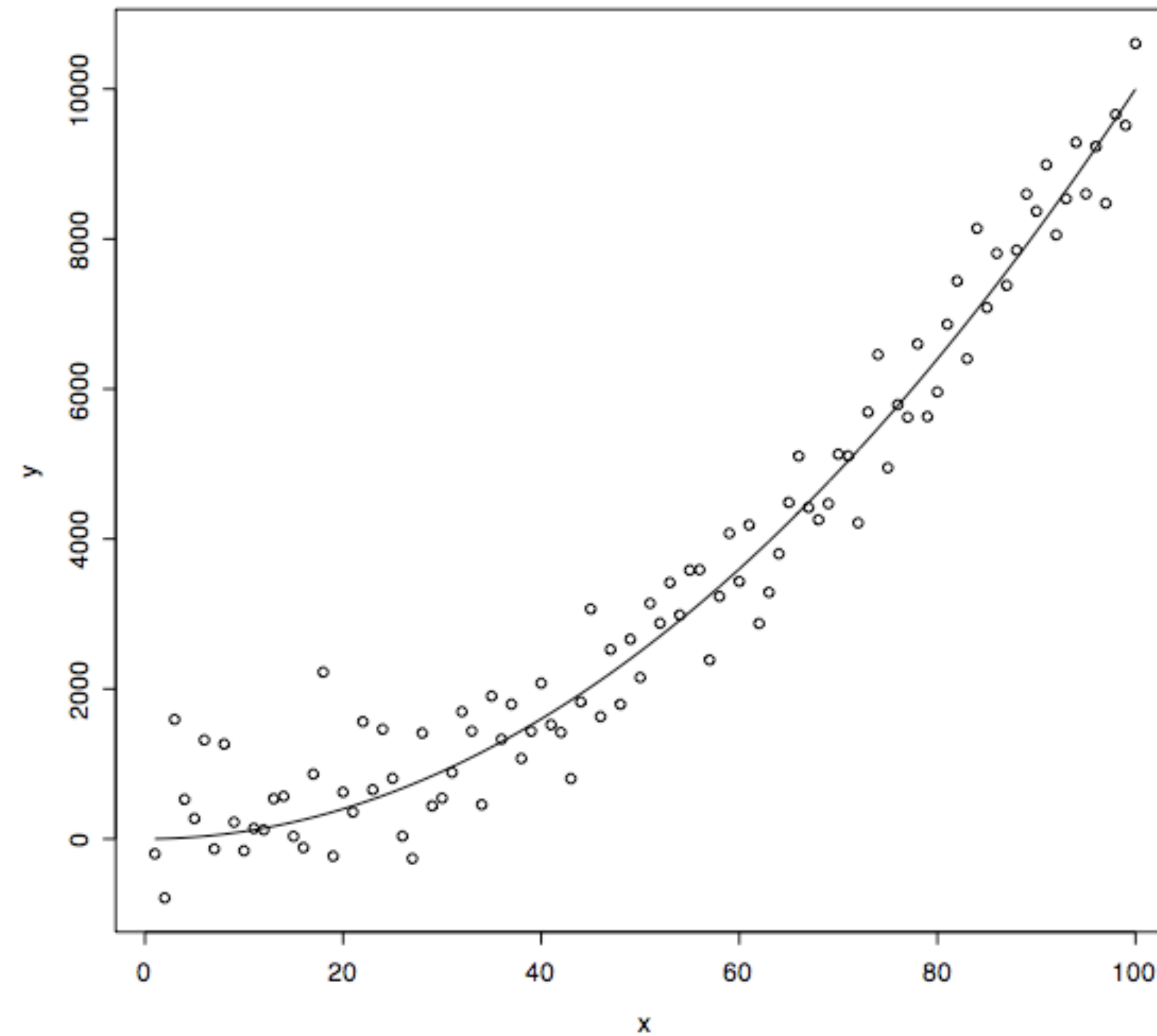
NO RELATIONSHIP



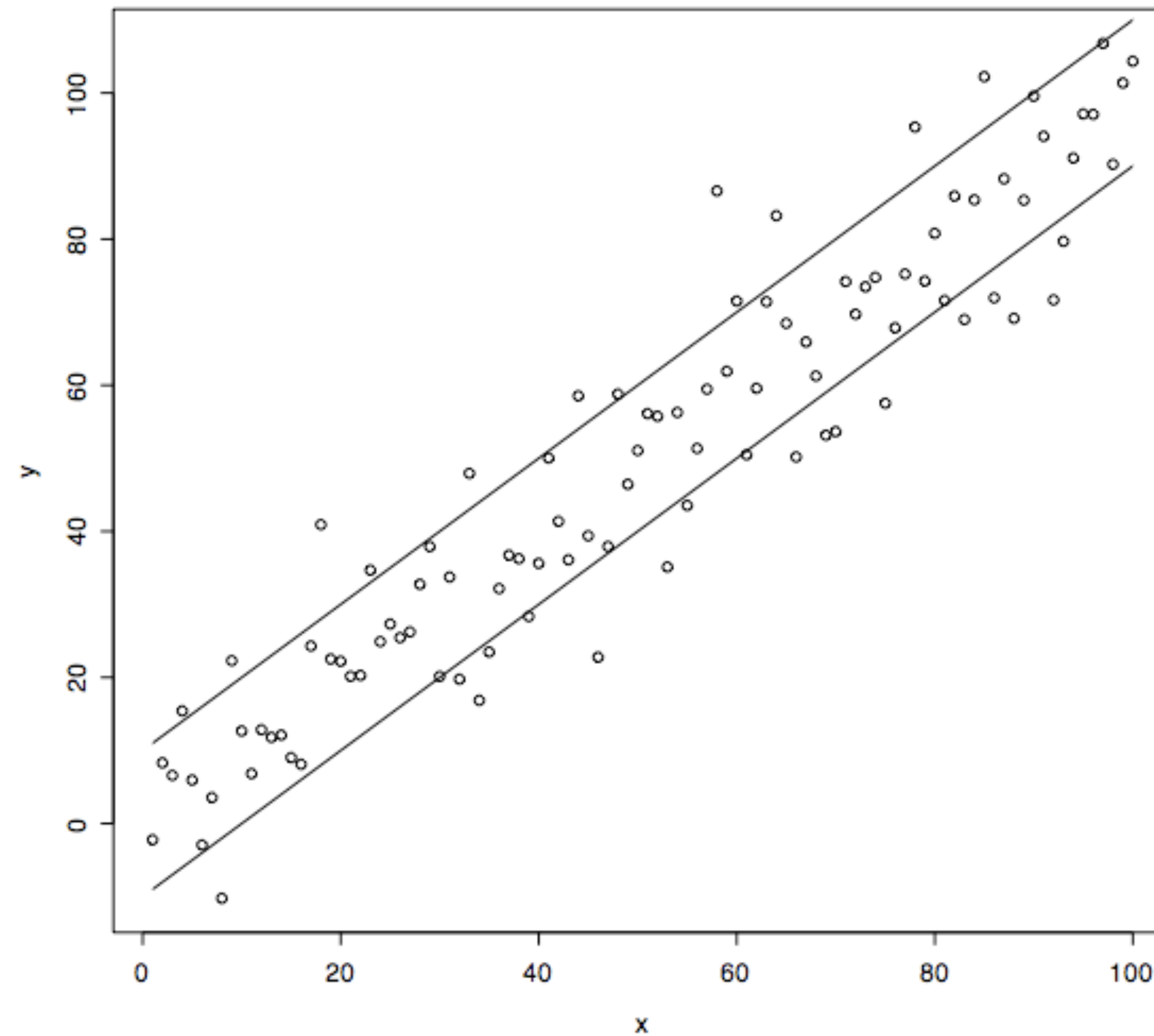
LINEAR RELATIONSHIP



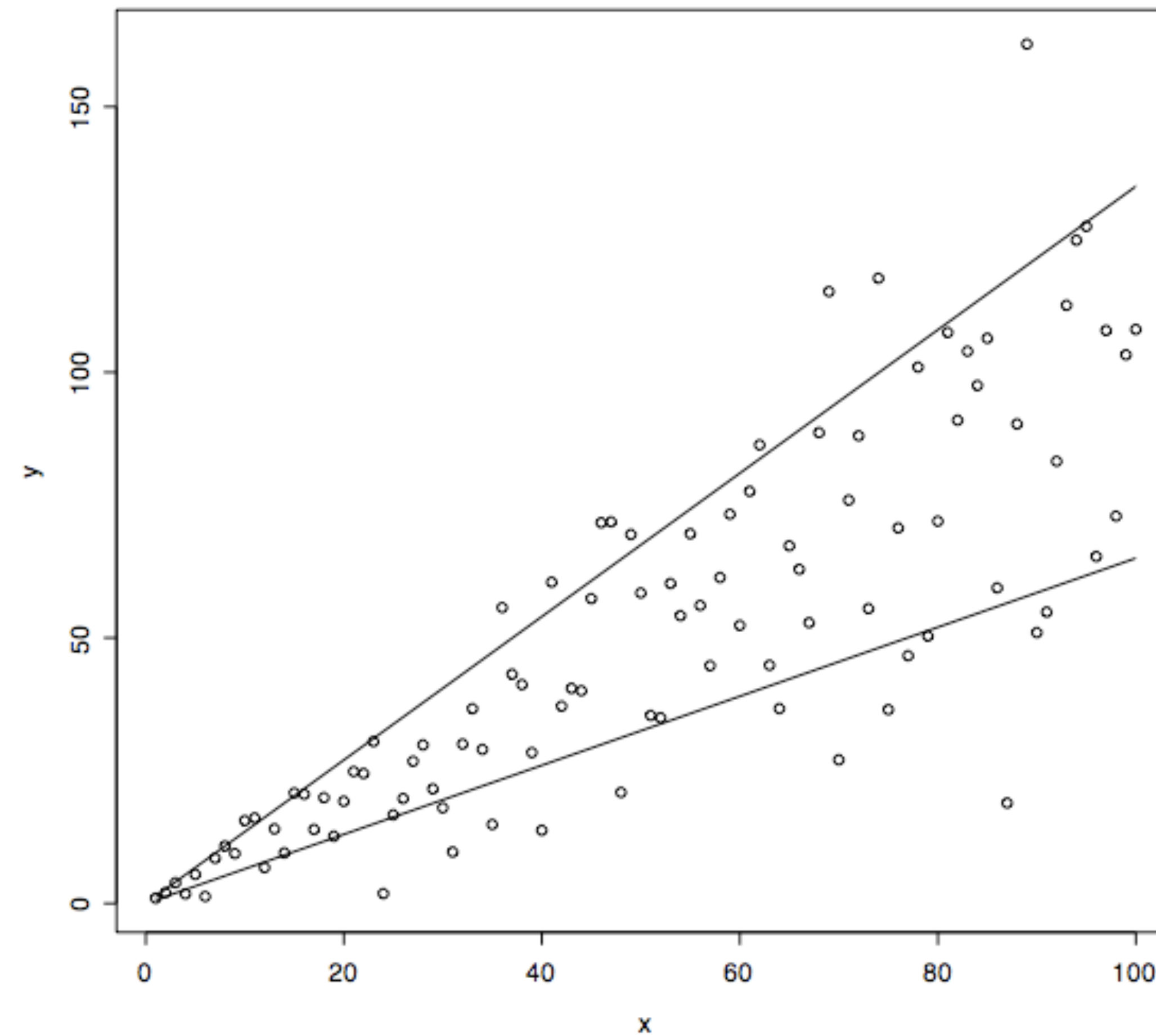
NON-LINEAR RELATIONSHIP



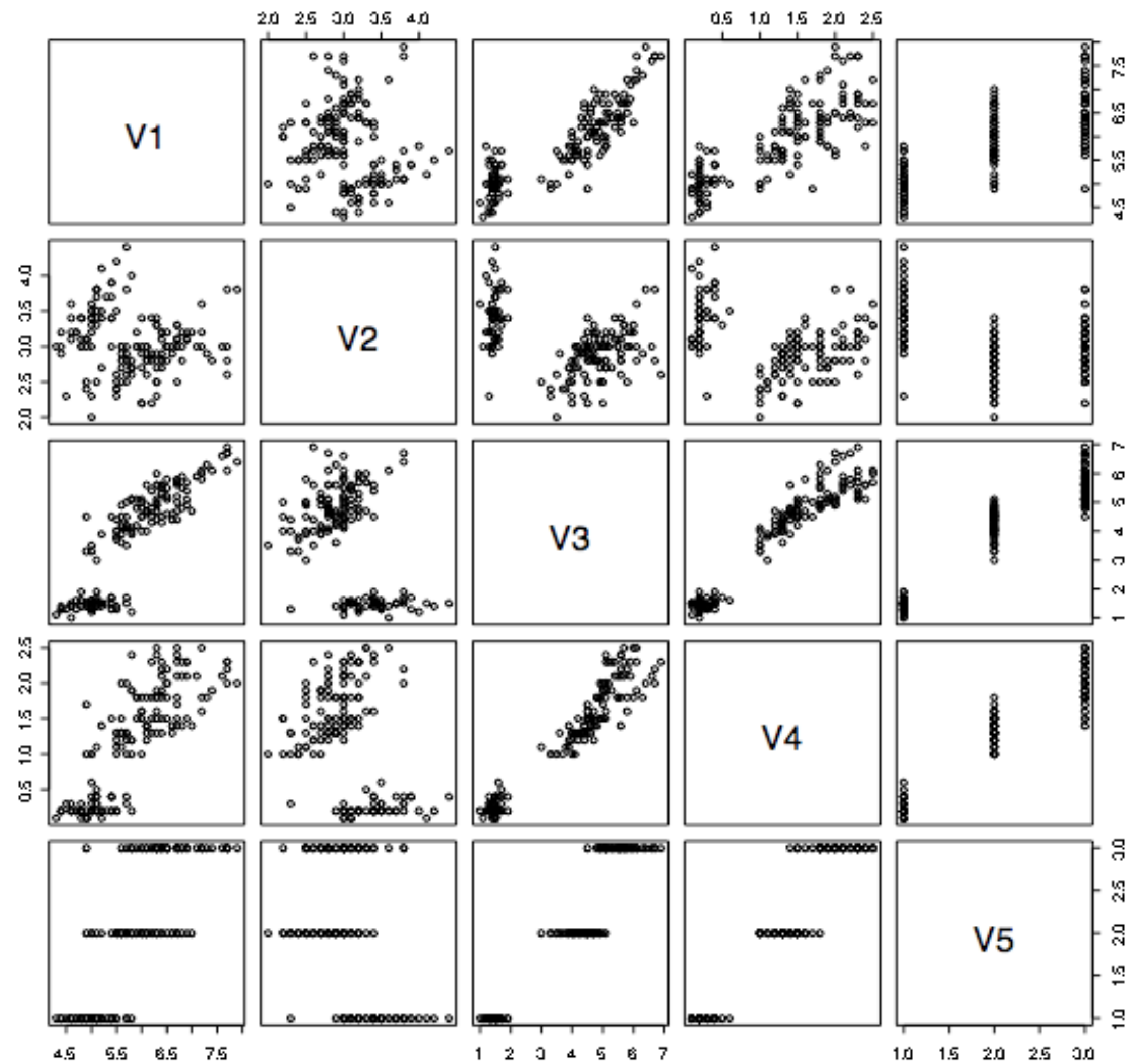
HOMOSKEDASTIC (EQUAL VARIANCE)



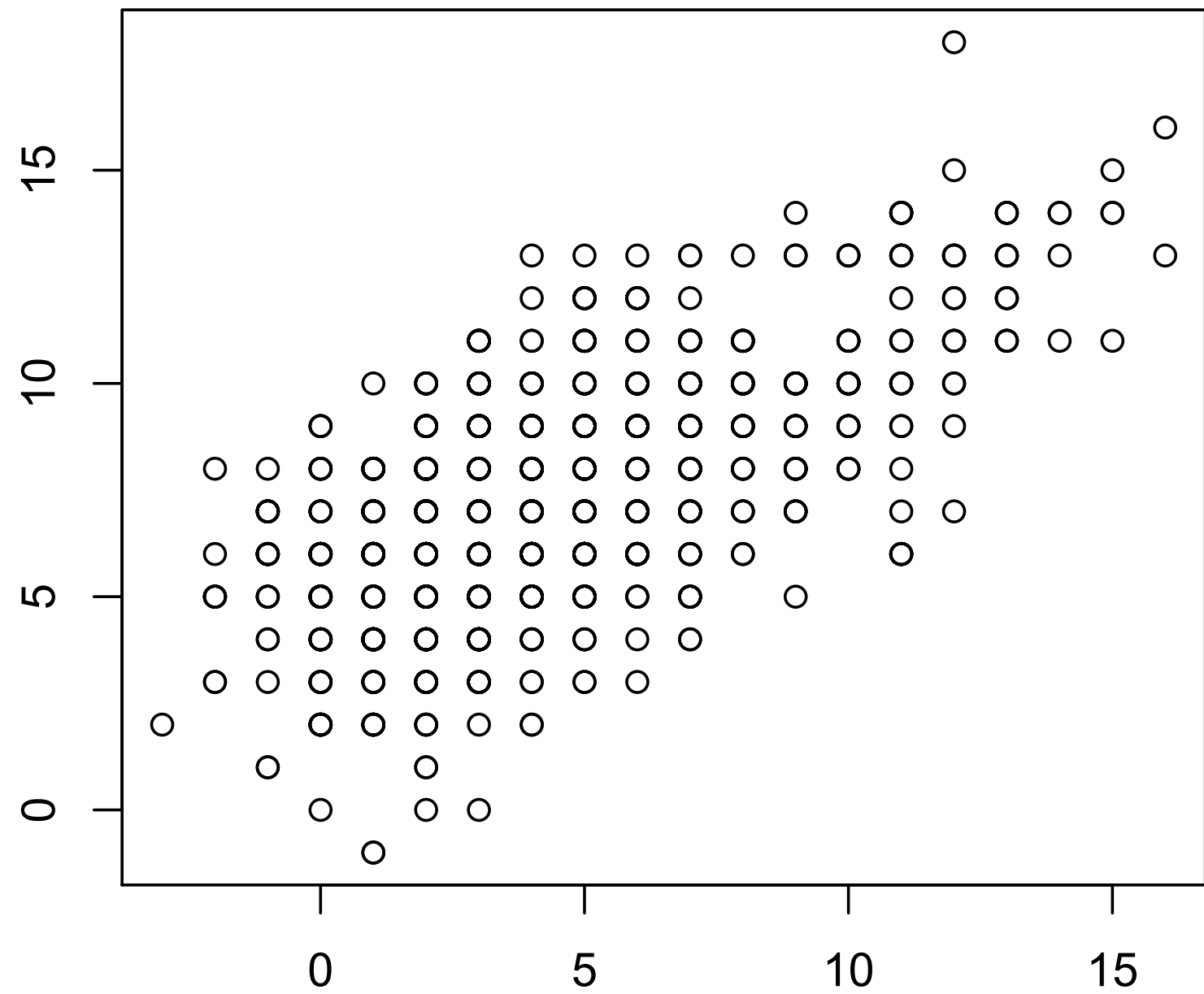
HETEROSKEDASTIC (UNEQUAL VARIANCE)



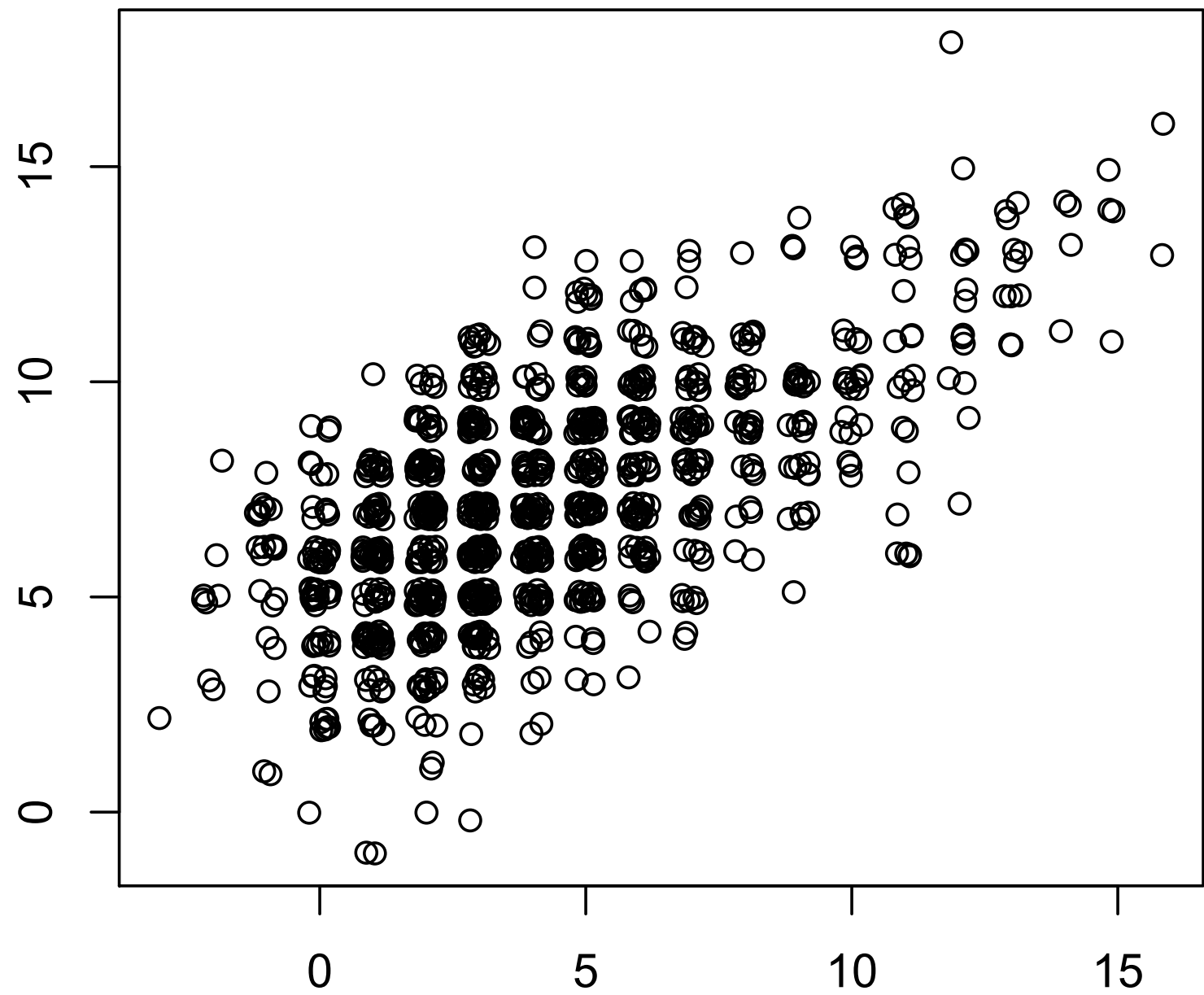
SCATTERPLOT MATRIX



SCATTERPLOT LIMITATIONS



Overprinting



Solution: Jitter points

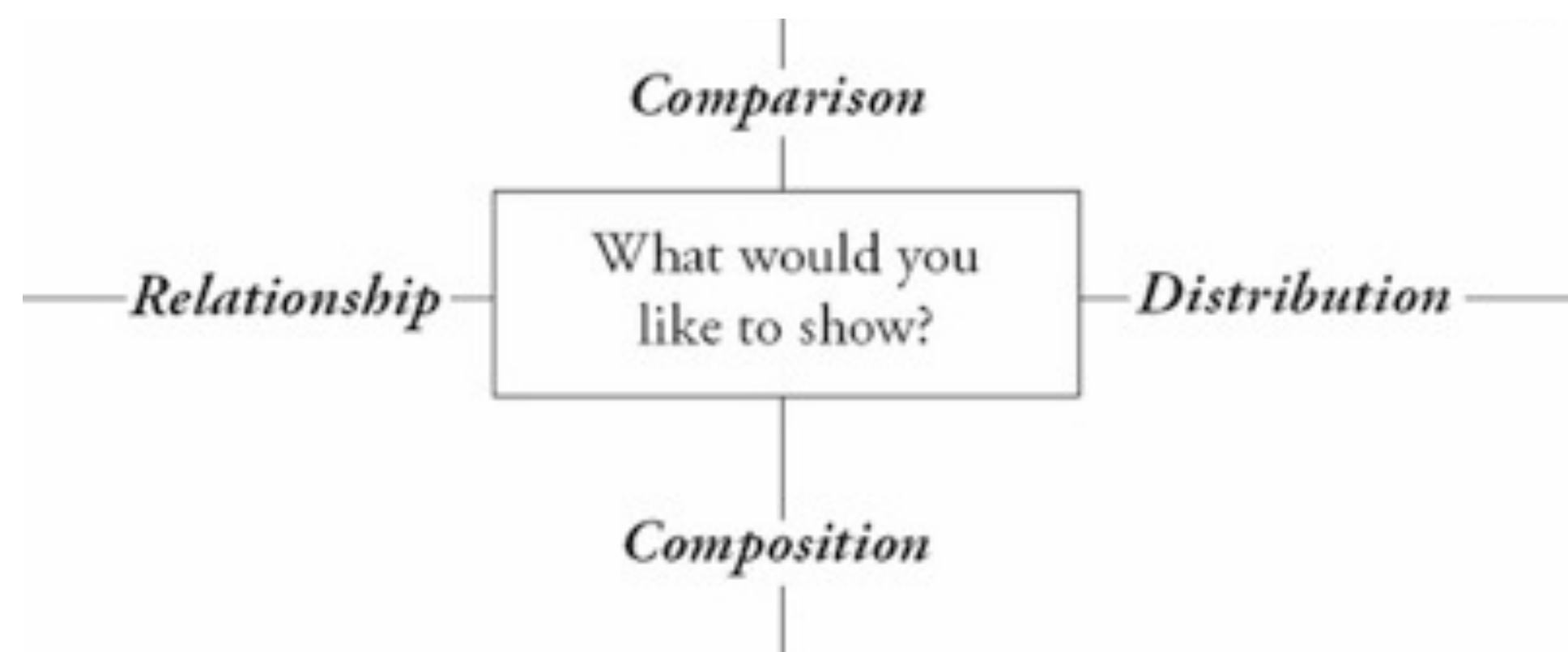
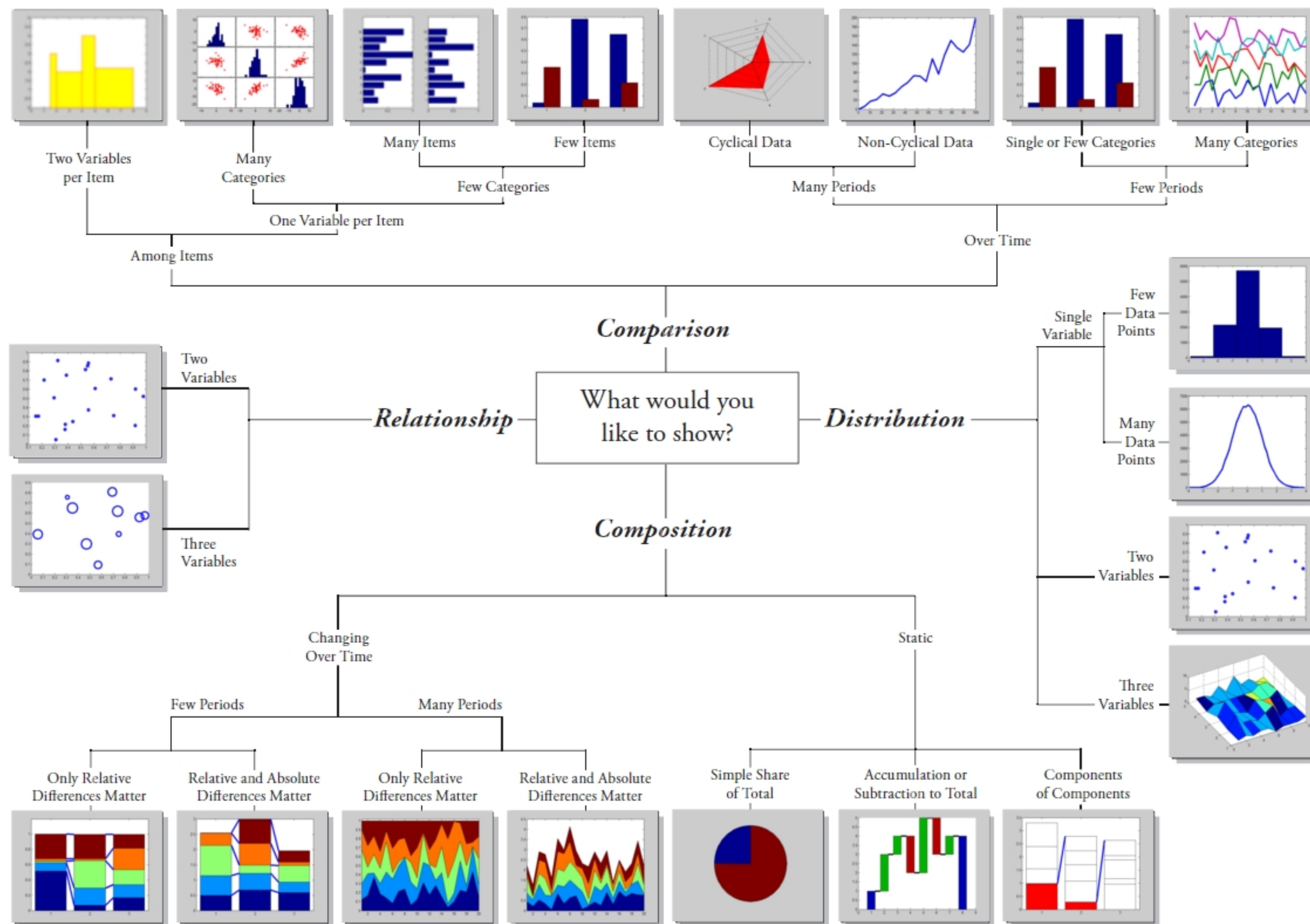


Chart Suggestions—A Thought-Starter



DIMENSIONALITY REDUCTION

DIMENSIONALITY REDUCTION

- ▶ Identify and describe the “dimensions” that underlie the data
 - ▶ May be more fundamental than those directly measured but hidden to the user
- ▶ Reduce dimensionality of modeling problem
 - ▶ Benefit is simplification, it reduces the number of variables you have to deal with in modeling
- ▶ Can identify set of variables with similar behavior
- ▶ Principal component analysis (PCA)

WHAT DIMENSION CAN BE DROPPED?

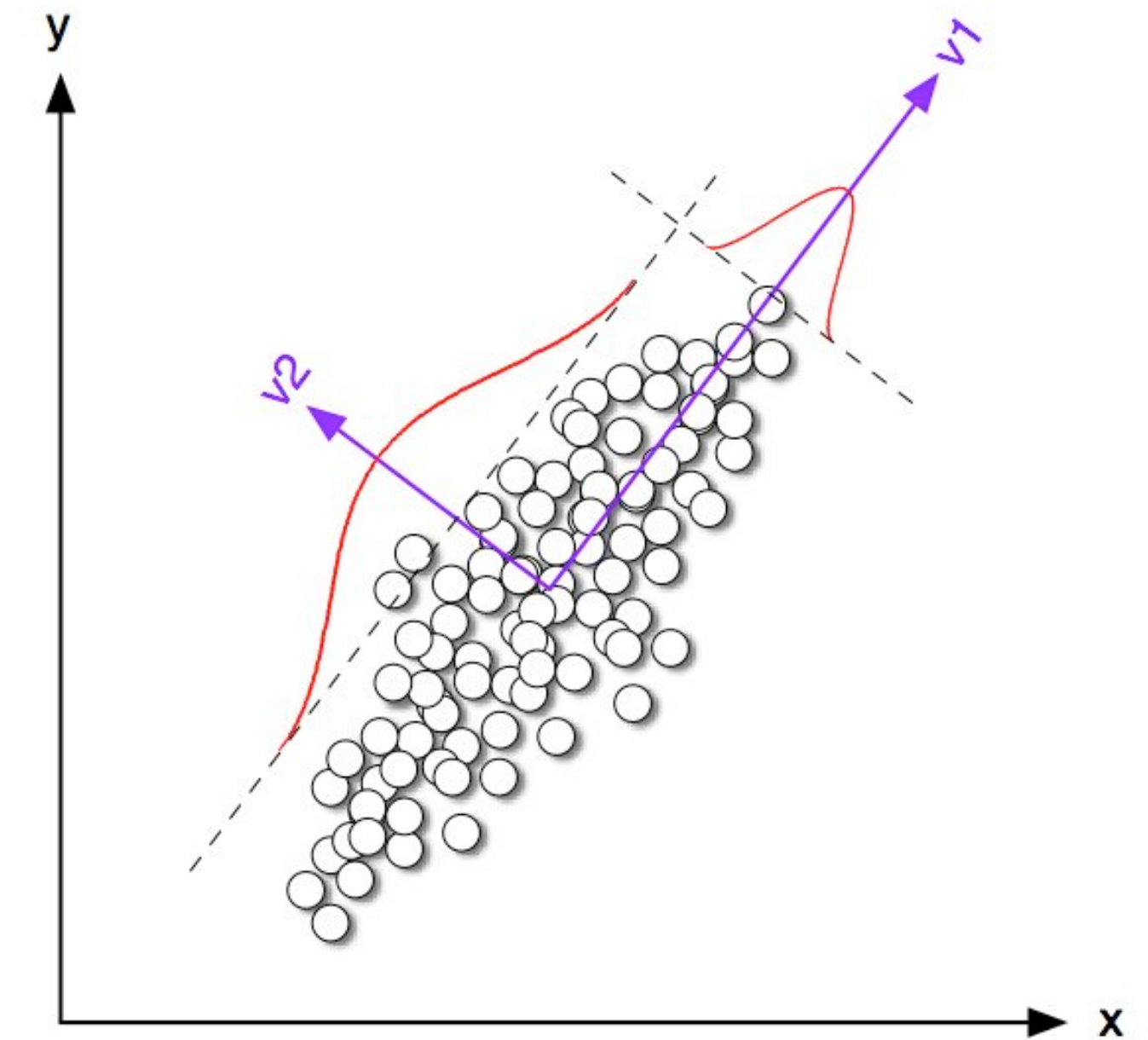
- ▶ Suppose we have a data matrix ***D*** of n rows and p columns (i.e., we have n data points, each data point is measured on p dimensions)
- ▶ If we want to decrease p , which dimensions can we drop?
 - ▶ Constant dimensions: $1, 1, \dots, 1$
 - ▶ Constant dimensions with some noise: $1.001, 0.998, \dots, 1.003$
 - ▶ Dimensions that is linearly dependent on other dimensions: $Z=aX+bY$

HIGH VARIANCE!

LOW COVARIANCE!

CHANGE OF BASIS

- ▶ But the dimension with highest variance may not necessarily be the dimension that we have measured
- ▶ Need change of basis such that:
 - ▶ The largest amount of variability of the data can be reflected by projecting the data to some basis vector in the new basis
 - ▶ After projecting the data to the new basis (or "new dimensions"), the covariances between new dimensions are low



PRINCIPLE COMPONENT ANALYSIS (PCA)

- ▶ Input: the $n \times p$ data matrix ***D***
- ▶ Preprocess ***D*** so that the mean of each dimension is 0, call this matrix ***X***

PRINCIPLE COMPONENT ANALYSIS (PCA)

- ▶ Each column vector of $\mathbf{Y}=\mathbf{XA}$ also has a mean of 0.

PRINCIPLE COMPONENT ANALYSIS (PCA)

- ▶ Input: the $n \times p$ data matrix $\mathbf{D}$
- ▶ Preprocess $\mathbf{D}$ so that the mean of each dimension is 0, call this matrix $\mathbf{X}$
- ▶ Goal: Find a $p \times p$ orthogonal transformation matrix $\mathbf{A}$ to conduct basis change, i.e., $\mathbf{Y} = \mathbf{XA}$, such that under the new basis, the covariances between the new dimensions are low
 - ▶ The $p \times p$ covariance matrix $\mathbf{Y}^T \mathbf{Y}$ is a diagonal matrix.

PRINCIPLE COMPONENT ANALYSIS (PCA)

$$\begin{aligned} Y^T Y &= (XA)^T (XA) \\ &= A^T X^T X A \end{aligned}$$

- ▶ Notice $\Sigma = X^T X$ is the covariance matrix under the current basis, it is a symmetric square matrix!
- ▶ So, we can conduct eigendecomposition for Σ

$$Y^T Y = A^T (Q \Lambda Q^T) A = (A^T Q) \Lambda (A^T Q)^T$$

PRINCIPLE COMPONENT ANALYSIS (PCA)

$$Y^T Y = A^T (Q \Lambda Q^T) A = (A^T Q) \Lambda (A^T Q)^T$$

- ▶ Let $A^T Q = I$, we get $A = (Q^{-1})^T = (Q^T)^T = Q$
 - ▶ By doing so, $Y^T Y = I \Lambda I = \Lambda$
- ▶ In other words, the transformation matrix A is Q , where each column is the eigenvector of the covariance matrix $\Sigma = X^T X$!
- ▶ The column vectors of A (or Q) are thus called the **principle component vectors**!